

Questions		Answers	Marks
3	(a)	$\lambda = \frac{v}{f}$ $= \frac{330}{1780}$ $= 0.185m$	A1
	(b)	$S_1D = \sqrt{12^2 + 4^2}$ $= 12.649 \text{ m}$ Path difference = $12.649 - 12$ $= \frac{0.649}{0.18539} \lambda$ $= 3.5\lambda$	M1 A1
	(c)	Since the two sources are in antiphase, <u>the waves meet at D exactly in phase</u> (Phase difference is 0 rad) and there is <u>constructive interference</u>. Hence a <u>maximum intensity is detected</u>.	B1 A1
	(d)	Since path difference remains unchanged and there are two more changes in sound intensity, Path difference = $(3.5 + 2) \lambda_{\text{new}}$ $= 5.5 \lambda_{\text{new}} \text{ (}\lambda_{\text{new}} \text{ is the new wavelength)}$ $5.5 \lambda_{\text{new}} = 3.5 \lambda$ $5.5\left(\frac{v}{f_{\text{new}}}\right) = 3.5(0.18539)$ $f_{\text{new}} = 2797 \text{ Hz}$ $= 2800 \text{ Hz (3s.f.)}$	B1 C1 A1